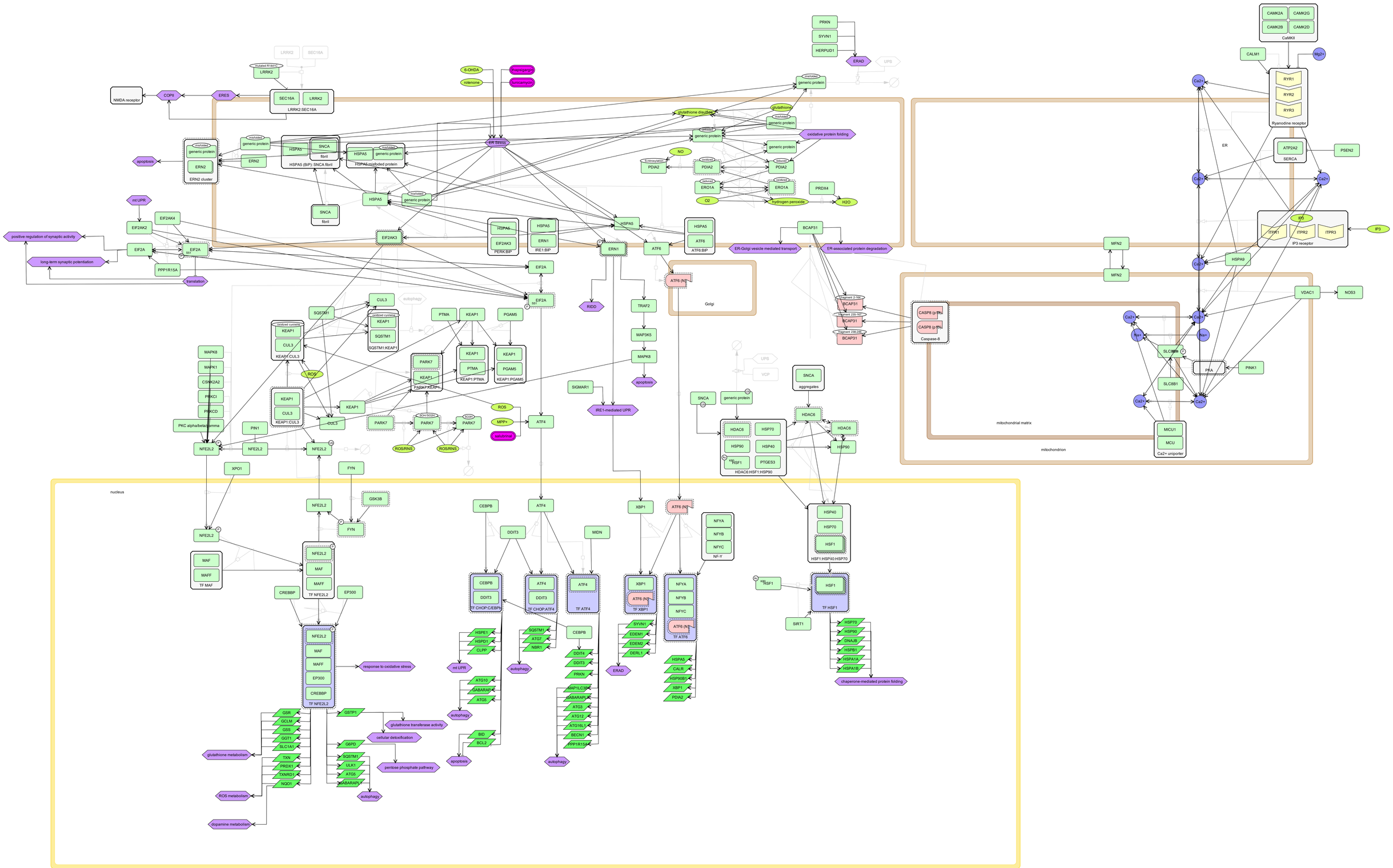

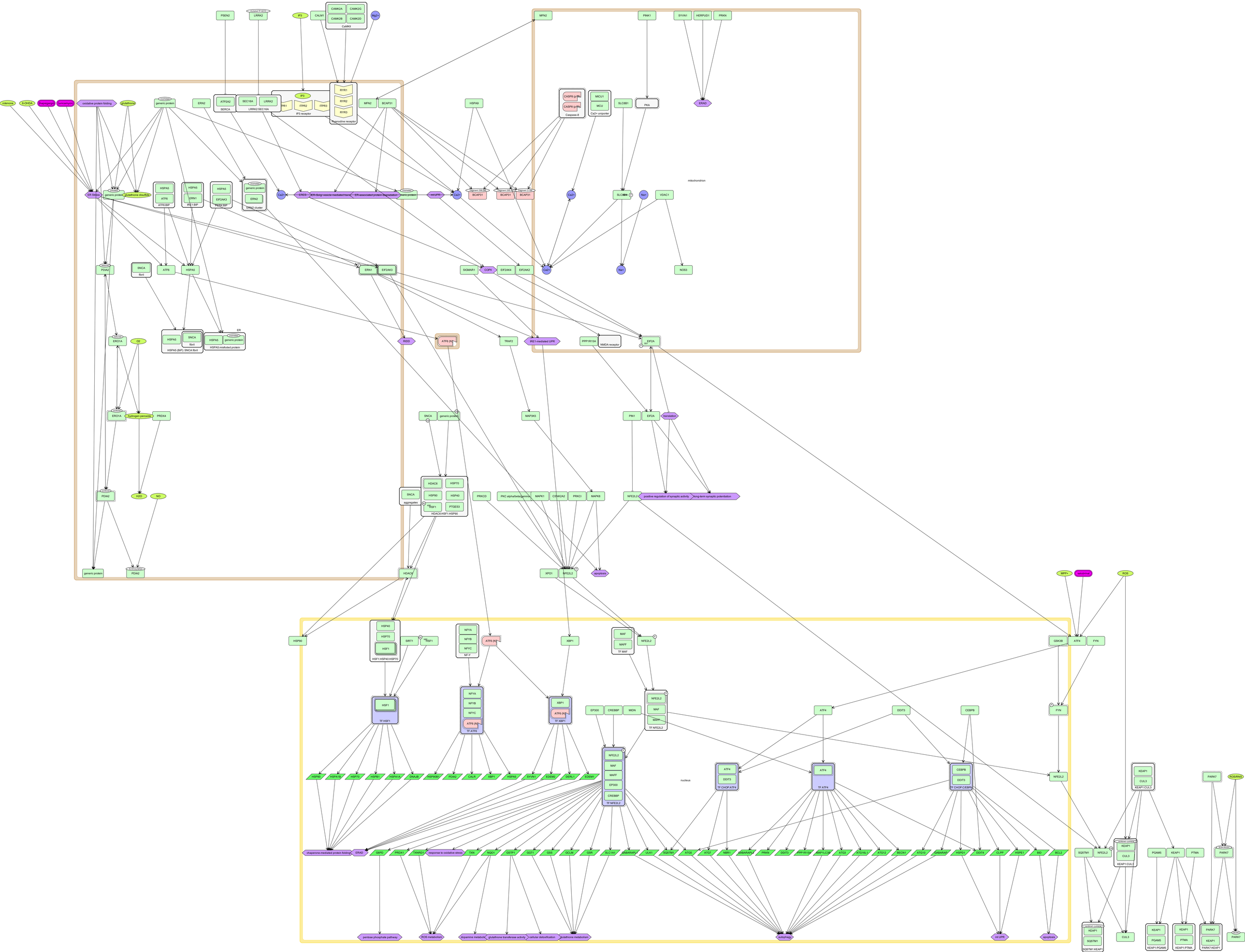

LRRK2
SEC16A
mutated R1441C
LRRK2
SEC16A
LRRK2
LRRK2:SEC16A
NMDA receptor
COPII
ERES
6-OHDA
rotenone
thapsigargin
tunicamycin
PRKN
SYVN1
HERPUD1
ERAD
UPS
misfolded
generic protein
glutathione disulfide
glutathione
misfolded
generic protein
oxidative protein folding
unfolded
generic protein
generic protein
NO
S-nitrosylation
PDIA2
oxidized
PDIA2
reduced
PDIA2
reduced
ERO1A
oxidized
ERO1A
PRDX4
O2
hydrogen peroxide
H2O
misfolded
generic protein
ERN2
ERN2 cluster
apoptosis
misfolded
generic protein
ERN2
HSPA5
SNCA
fibril
HSPA5 (BiP): SNCA fibril
HSPA5
misfolded
generic protein
HSPA5:misfoded protein
ER Stress
HSPA5
misfolded
generic protein
SNCA
fibril
mt UPR
EIF2AK4
EIF2AK2
EIF2AK3
HSPA5
EIF2AK3
PERK:BIP
HSPA5
ERN1
IRE1:BIP
HSPA5
HSPA5
ATF6
ATF6:BIP
ER-Golgi vesicle mediated transport
BCAP31
ER-associated protein degradation
positive regulation of synaptic activity
EIF2A
EIF2A
S51
P
long-term synaptic potentiation
PPP1R15A
translation
ERN1
P
ATF6
ATF6 (N)
Golgi
EIF2A
EIF2A
S51
P
RIDD
TRAF2
MAP3K5
MAPK8
apoptosis
SIGMAR1
IRE1-mediated UPR
fragment 2-164
BCAP31
fragment 235-167
BCAP31
fragment 238-246
BCAP31
CASP8 (p18)
CASP8 (p10)
Caspase-8
SQSTM1
CUL3
autophagy
oxidized cysteine
KEAP1
SQSTM1
SQSTM1:KEAP1
oxidized cysteine
KEAP1
CUL3
KEAP1:CUL3
PTMA
KEAP1
PGAM5
KEAP1
PTMA
KEAP1:PTMA
KEAP1
PGAM5
KEAP1:PGAM5
PARK7
KEAP1
PARK7:KEAP1
ROS
KEAP1
CUL3
KEAP1:CUL3
KEAP1
CUL3
PARK7
SOH/SO2H
PARK7
SO3H
PARK7
ROS/RNS
ROS/RNS
ROS
MPP+
salubrinal
ATF4
MAPK8
MAPK1
CSNK2A2
PRKCI
PRKCD
PKC alpha/beta/gamma
PIN1
NFE2L2
P
NFE2L2
NFE2L2
Ub
XPO1
FYN
UPS
VCP
generic protein
Ub
SNCA
Ub
SNCA
aggregates
HDAC6
HSP70
HSP90
HSP40
Ac
K80
HSF1
PTGES3
HDAC6:HSF1:HSP90
HDAC6
HDAC6
HSP90
nucleus
NFE2L2
GSK3B
P
FYN
NFE2L2
P
MAF
MAFF
TF MAF
NFE2L2
P
MAF
MAFF
TF NFE2L2
CREBBP
EP300
NFE2L2
P
MAF
MAFF
EP300
CREBBP
TF NFE2L2
response to oxidative stress
GSR
GCLM
GSS
GGT1
SLC1A1
glutathione metabolism
TXN
PRDX1
TXNRD1
NQO1
ROS metabolism
dopamine metabolism
GSTP1
glutathione transferase activity
cellular detoxification
G6PD
pentose phosphate pathway
SQSTM1
ULK1
ATG5
GABARAPL1
autophagy
CEBPB
ATF4
DDIT3
MIDN
CEBPB
DDIT3
TF CHOP:C/EBPb
ATF4
DDIT3
TF CHOP:ATF4
ATF4
TF ATF4
HSPE1
HSPD1
CLPP
mt UPR
SQSTM1
ATG7
NBR1
autophagy
CEBPB
ATG10
GABARAP
ATG5
autophagy
BID
BCL2
apoptosis
DDIT4
DDIT3
PRKN
MAP1LC3B
GABARAPL2
ATG3
ATG12
ATG16L1
BECN1
PPP1R15A
autophagy
XBP1
ATF6 (N)
NFYA
NFYB
NFYC
NF-Y
XBP1
ATF6 (N)
TF XBP1
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
SYVN1
EDEM1
EDEM2
DERL1
ERAD
HSPA5
CALR
HSP90B1
XBP1
PDIA2
HSP40
HSP70
HSF1
HSF1:HSP40:HSP70
Ac
K80
HSF1
HSF1
TF HSF1
SIRT1
HSP70
HSP90
DNAJB
HSPB1
HSPA1A
HSPA1B
chaperone-mediated protein folding
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
CALM1
Mg2+
Ca2+
RYR1
RYR2
RYR3
Ryanodine receptor
ER
ATP2A2
SERCA
PSEN2
Ca2+
Ca2+
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
IP3
MFN2
MFN2
Ca2+
HSPA9
VDAC1
NOS3
Ca2+
Ca2+
Na+
Na+
SLC8B1
P
PKA
PINK1
SLC8B1
Ca2+
Ca2+
mitochondrial matrix
MICU1
MCU
Ca2+ uniporter
mitochondrion

PSEN2
mutated R1441C
LRRK2
IP3
CALM1
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
Mg2+
MFN2
PINK1
SYVN1
HERPUD1
PRKN
rotenone
6-OHDA
thapsigargin
tunicamycin
oxidative protein folding
glutathione
misfolded
generic protein
ERN2
ATP2A2
SERCA
SEC16A
LRRK2
LRRK2:SEC16A
ITPR1
ITPR2
ITPR3
IP3 receptor
RYR1
RYR2
RYR3
Ryanodine receptor
MFN2
BCAP31
HSPA9
CASP8 (p18)
CASP8 (p10)
Caspase-8
MICU1
MCU
Ca2+ uniporter
SLC8B1
PKA
ERAD
ER Stress
unfolded
generic protein
glutathione disulfide
HSPA5
ATF6
ATF6:BIP
HSPA5
ERN1
IRE1:BIP
HSPA5
EIF2AK3
PERK:BIP
misfolded
generic protein
ERN2
ERN2 cluster
Ca2+
ERES
ER-Golgi vesicle mediated trans
ER-associated protein degradation
misfolded
generic protein
mt UPR
Ca2+
fragment 238-246
BCAP31
fragment 235-167
BCAP31
fragment 2-164
BCAP31
Ca2+
SLC8B1
P
Na+
VDAC1
mitochondrion
reduced
PDIA2
SNCA
fibril
ATF6
HSPA5
P
ERN1
EIF2AK3
SIGMAR1
COPII
EIF2AK4
EIF2AK2
Ca2+
Na+
NOS3
HSPA5
SNCA
fibril
HSPA5 (BiP): SNCA fibril
ER
HSPA5
misfolded
generic protein
HSPA5:misfoded protein
reduced
ERO1A
O2
RIDD
ATF6 (N)
TRAF2
IRE1-mediated UPR
PPP1R15A
NMDA receptor
S51
P
EIF2A
oxidized
ERO1A
hydrogen peroxide
PRDX4
SNCA
Ub
Ub
generic protein
MAP3K5
PIN1
EIF2A
translation
HDAC6
HSP70
HSP90
HSP40
Ac
K80
HSF1
PTGES3
HDAC6:HSF1:HSP90
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PRKCI
MAPK8
NFE2L2
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long-term synaptic potentiation
oxidized
PDIA2
H2O
NO
generic protein
S-nitrosylation
PDIA2
HDAC6
XPO1
P
NFE2L2
apoptosis
MPP+
salubrinal
ROS
HSP90
HSP40
HSP70
HSF1
HSF1:HSP40:HSP70
SIRT1
Ac
K80
HSF1
NFYA
NFYB
NFYC
NF-Y
ATF6 (N)
XBP1
MAF
MAFF
TF MAF
P
NFE2L2
GSK3B
ATF4
FYN
HSF1
TF HSF1
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
XBP1
ATF6 (N)
TF XBP1
P
NFE2L2
MAF
MAFF
TF NFE2L2
EP300
CREBBP
MIDN
ATF4
DDIT3
CEBPB
P
FYN
P
NFE2L2
MAF
MAFF
EP300
CREBBP
TF NFE2L2
nucleus
ATF4
DDIT3
TF CHOP:ATF4
ATF4
TF ATF4
CEBPB
DDIT3
TF CHOP:C/EBPb
NFE2L2
KEAP1
CUL3
KEAP1:CUL3
PARK7
ROS/RNS
HSP90
HSPA1B
HSP70
HSPB1
HSPA1A
DNAJB
HSP90B1
PDIA2
CALR
XBP1
HSPA5
SYVN1
EDEM2
DERL1
EDEM1
chaperone-mediated protein folding
ERAD
G6PD
PRDX1
TXNRD1
response to oxidative stress
TXN
NQO1
GSTP1
GGT1
GSS
GCLM
GSR
SLC1A1
GABARAPL1
ULK1
SQSTM1
ATG5
ATG7
NBR1
GABARAPL2
PRKN
DDIT3
PPP1R15A
MAP1LC3B
ATG3
ATG16L1
ATG12
BECN1
ATG10
GABARAP
HSPD1
DDIT4
CLPP
HSPE1
BID
BCL2
SQSTM1
Ub
NFE2L2
oxidized cysteine
KEAP1
CUL3
KEAP1:CUL3
PGAM5
KEAP1
PTMA
SOH/SO2H
PARK7
pentose phosphate pathway
ROS metabolism
dopamine metabolic
glutathione transferase activity
cellular detoxification
glutathione metabolism
autophagy
mt UPR
apoptosis
oxidized cysteine
KEAP1
SQSTM1
SQSTM1:KEAP1
CUL3
KEAP1
PGAM5
KEAP1:PGAM5
KEAP1
PTMA
KEAP1:PTMA
PARK7
KEAP1
PARK7:KEAP1
SO3H
PARK7